Computational Materials Science 104 (2015) 138–142

Contents lists available at ScienceDirect

Computational Materials Science

journal homepage: www.elsevier.com/locate/commatsci

Thermodynamic assessment of the Mg–Pr system

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ARTICLE INFO

Article history:
Received 9 September 2014
Received in revised form 20 November 2014
Accepted 21 November 2014
Available online 20 April 2015

Keywords:
Calphad-assessment
Magnesium alloys
Rare earths
Mg–Pr

ABSTRACT

Phase equilibria and thermodynamic data of the Mg–Pr system were assessed using the Calphad-method based on a critical review of the experimental data. A simple subregular solution model for the liquid phase is sufficient for a quantitative thermodynamic description and short-range order need not to be assumed in the Mg–Pr melt. A quantitative description of the phase equilibria in the important Mg-rich side is provided and some aspects on the Pr-rich side are also improved in the present description, compared to previous work, even though simpler models have been used.

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1. Introduction

Alloying additions of rare earths (RE) to Mg alloys are of industrial relevance to improve the mechanical properties of these light weight alloys in transportation and other applications. Such alloys are also used in extrusion processes [1], which are more challenging for Mg alloys because of the limited plasticity of hcp-Mg compared to fcc-Al alloys. Other advanced applications exploit the relatively large and retrograde solubility of some RE elements in the (Mg) solid solution in ageing heat treatment for solution and precipitation strengthening, improved creep resistance, but also for improved sheet formability [2].

One of the relevant RE components is praseodymium. To better understand the phase formation in such alloys and to focus alloy and process development quantitative knowledge of the Mg–Pr phase diagram is essential, as well as a validated thermodynamic description for use in more detailed calculations, also in multicomponent Mg alloys.

The first thermodynamic assessment of the Mg–Pr system by Guo and Du [3] was later revised by that group [4]. However, their assessments show significant deviations from experimental data in the important Mg-rich side. That region was better described in the later work by Kang et al. [5]. However, significant deviations exist in the Pr-rich side, as also mentioned by Kang et al. [5]. Another difficulty is the description of the liquid phase using the modified quasichemical model (MQM) [5]. Firstly, significant short-range order was assumed in the Mg–Pr melt in that MQM description without any direct experimental evidence. Secondly, the MQM is available only in the Factsage software package [6], so it is hard to reproduce using other generally used softwares such as Pandat [7] or Thermocalc [8].

The purpose of this work is to show that the simple subregular solution model for the liquid phase, available in all software packages, is sufficient for a quantitative thermodynamic description of the Mg–Pr system and that short-range order need not to be assumed. In addition, the present re-assessment of this system aims at a quantitative description of the important Mg-rich side.

2. Summary of experimental literature data

Table 1 summarizes crystal structure data of all Mg–Pr solid phases. The reported homogeneity ranges are also given. It should be noticed that some composition range data are questionable, which will be discussed later.

Canneri and Rossi [9] determined the enthalpies of formation of PrMg and $PrMg_3$ phases by means of acid-solution calorimetry, which is not a very accurate method. Ogren et al. [10] has measured the vapor pressure of the Mg–Pr alloys by the Knudsen effusion method. However, our previous work on the Mg–Gd system [11], indicated that these data [10] are questionable. Thus, in the present work, the data from [10] were not used.

The Mg–Pr phase equilibria have been investigated by several groups of researchers [9,12–16]. Canneri and Rossi [9] investigated the Mg–Pr system by thermal analysis and metallographic methods. The measured melting points of Mg (661 °C) and Pr (932 °C) show that the starting Mg material was of low purity. Thus, in the present work, the phase equilibrium data [9] was not

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http://dx.doi.org/10.1016/j.commatsci.2014.11.050
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Table 1
Solid phases of the Mg–Pr system.

Phase	Pearson symbol Space group Prototype	Lattice parameters (nm)	Homogeneity range (at.% Pr)	Reference to homogeneity range
Mg	hP2 $P6_3/mmc$ Mg	a = 0.32094 c = 0.52107	0–0.297	[18] [2003Rok]
$PrMg_{12}$	tI26 $I4/mmm$ $Mn_{12}Th$	a = 1.034 c = 0.598	7.7 8.2–9.9	[15] [1993Sac] [18] [2003Rok]
Pr_5Mg_{41}	tI92 $I4/m$ $Mg_{41}Ce_5$	a = 0.1475 c = 0.1041	10.9 about 10	[15] [1993Sac] [18] [2003Rok]
$PrMg_3$	cF16 Fm-3m BiF_3	a = 0.7430 c = 0.7411	19–25 ?–25	[15] [1993Sac] [18] [2003Rok]
$PrMg_2$[a]	cF24 Fd-3m O2 Cu_2Mg	a = 0.8689	33.3 33.33	[15] [1993Sac] [18][2003Rok]
PrMg	cP2 Pm-3m CsCl	a = 0.3898	~47 to ~53 50	[15] [1993Sac] [18] [2003Rok]
βPr[a]	cI2 Im-3m W	a = 0.413	68-100	[15] [1993Sac]
αPr	hP4 $P6_3/mmc$ Nd	a = 0.36721 b = 1.18326	92–100 90–100	[13] [1965Jos] [15] [1993Sac]

Lattice parameters are from [15], space group from [22].
[a] Phase is stable at high temperature only.

considered. Park and Wyman [12] and Bochvar [14] measured the solubility of Pr in solid Mg by means of the X-ray method and resistivity method. However, their data are inconsistent. The reported solubility of Pr in solid Mg at 540 °C is 0.07 at.% [12] which is far away from the value of 0.27 at.% Pr [14]. Due to the fact that the solubility of Pr in (Mg) is very small, the change of the Mg lattice parameter turns out to be quite small while the change of electrical resistivity is substantial. Thus, the resistivity data [14] is much more reliable in this case. Bochvar [14] also measured the liquidus and invariant reactions in the Mg-rich corner by means of thermal analysis. Joseph and Gschneidner [13] measured the solid solubility of Mg in αPr using X-ray parametric technique. The Mg–Pr system was assessed by Nayeb-Hashemi and Clark [17] considering all the previous experimental data. The most recent experimental research is done by Saccone et al. [15,16]. Saccone et al. [15] re-investigated the Mg–Pr phase equilibrium by means of X-ray powder diffraction, optical and scanning electron microscopy, electron probe microanalysis and differential thermal analysis. The melting behavior of the Mg–Pr system was systematically investigated. Based on the above experimental information, the Mg–Pr system was assessed by Ref. [18]. The conclusions are same as given by Ref. [15] except for the Mg-rich part. The phase equilibrium data in the Mg-rich corner by Ref. [18] are mostly based on Bochvar's data [14]. By means of Smith thermal analysis, Saccone et al. [16] re-determined the solvus lines in the Pr-rich region and temperatures of the invariant reactions of the Mg-rich region.

3. Thermodynamic modeling

3.1. Pure elements

The Gibbs energy of the pure element i in phase ϕ is expressed in this work by the following equation:

$$G_i^{0,\phi}(T) = G_i^{\phi}(T) - H_i^{SER} = A + BT + CT\ln T + DT^2 + ET^3 + FT^{-1} + GT^7 + HT^{-9} + IT^4 \quad (1)$$

where H_i^{SER} is the molar enthalpy of the element (i) at 298.15 K and 1 bar in its standard element reference (SER) state. T is the absolute temperature. The "0" in $G_i^{0,\phi}(T)$ defines the so-called SER-reference state, which is, thus, transferred to all phases in the subsequent equations. The Gibbs energy functions for Mg and Pr above 298 K are taken from the SGTE (v5.1) database (Scientific Group Thermodata Europe), which is identical to the printed version [19].

3.2. Solution phases

The Gibbs energy for the solution phases ϕ (αPr, βPr, (Mg), and liquid) are described by the substitutional solution model with a Redlich–Kister polynomial [20] for the excess term in the subregular approximation as follows:

$$G^{\phi} = x_{Mg}\cdot G_{Mg}^{0,\phi} + x_{Pr}\cdot G_{Pr}^{0,\phi} + R\cdot T\cdot(x_{Mg}\cdot\ln x_{Mg} + x_{Pr}\cdot\ln x_{Pr}) + x_{Mg}\cdot x_{Pr}\left[L_{Mg,Pr}^{0,\phi} + L_{Mg,Pr}^{1,\phi}(x_{Mg} - x_{Pr})\right] \quad (2)$$

where x_{Mg} and x_{Pr} represent the mole fractions of elements Mg and Pr, respectively. R is the gas constant (R = 8.3145 J mol^{-1} K^{-1}) and $L_{Mg,Pr}^{v,Liquid}$ is the Redlich–Kister parameter representing the interaction of order v between components Mg and Pr in the liquid phase. The second order (subregular, v = 0 and 1) is used for the liquid and βPr phase only, whereas αPr and (Mg) are described as simple regular solutions (v = 0). A linear temperature dependence was used for the interaction parameter $L_{Mg,Pr}^{v,Liquid}$:

$$L_{Mg,Pr}^{v,Liquid} = A_v + B_v\cdot T \quad (3)$$

3.3. Intermetallic phases

In the present work most intermetallic phases were modeled as simple stoichiometric compounds. In total, these are $PrMg_{12}$, Pr_5Mg_{41}, $PrMg_3$, and $PrMg_2$. The potential solution range of $PrMg_3$, is discussed later. For example, the Gibbs energy of Pr_5Mg_{41} is expressed with the following expression:

$$G^{Pr_5Mg_{41}} = 5\cdot G_{Pr}^{0,Dhcp} + 41\cdot G_{Mg}^{0,Hcp} + \Delta G_f^{Pr_5Mg_{41}} \quad (4)$$

in which and $\Delta G_f^{Pr_5Mg_{41}} = A + B\cdot T$ represents the Gibbs energy of formation of the compound Pr_5Mg_{41} from the components, as defined in Eq. (1), in a reference state floating with temperature. The parameters A and B are to be optimized.

Although experimental data about the homogeneity range of the PrMg phase may be questionable, the related Mg–Y system shows that there may be a small homogeneity range. Thus, in order to be consistent, a narrow solubility range was allowed for the PrMg phase and it was modeled with simple two-sublattice model (**Pr**, Mg)(**Mg**), where the first sublattice is mainly occupied by Pr, the second only by Mg. The Gibbs energy of PrMg per mol-formula can be expressed as the follows:

$$G^{PrMg} = G_{Pr:Mg}^{PrMg}\cdot y_{Pr} + G_{Mg:Mg}^{PrMg}\cdot y_{Mg} + RT\cdot(y_{Pr}\ln y_{Pr} + y_{Mg}\ln y_{Mg}) \quad (5)$$

where y_{Mg} and y_{Pr} are the site fractions of the constituents Mg and Pr in the first sublattice, respectively. $G_{Pr:Mg}^{PrMg}$ is the Gibbs energy of the ideal compound PrMg, expressed in analogy to Eq. (4). The second end-member parameter of the sublattice solution range is $G_{Mg:Mg}^{PrMg}$, the Gibbs energy of pure Mg in the structure of PrMg, which is assigned a value well above that of stable hcp-Mg. No interaction parameter is necessary in the PrMg phase.

3.4. Parameter optimization

The parameter optimization was performed using all the accepted experimental thermodynamic and phase equilibrium data and is based mainly on the following data. The phase equilibria in the Mg-rich corner from [18], the phase equilibria for the solid solubility of Mg in αPr [13], all the remaining phase equilibria from [15], and the enthalpies of formation for PrMg and $PrMg_3$ phases [9]. All thermodynamic and phase diagram calculations were performed using both of the independent software packages Pandat (version 8.2) [7] and Thermocalc (version S) [8], to ensure the portability of the present dataset. The use of Pandat is also advisable to ensure the true stability in the phase diagrams by global minimization of Gibbs energy in all types of calculations.

4. Results and discussion

The parameters of the Gibbs energy functions determined in this work are compiled in Table 2. The parameter of $PrMg_2$ is adopted from Guo et al. [4]. During the assessment procedure it was found that the $PrMg_2$ phase-parameter from Guo et al. [4] can describe the corresponding experimental data well. That is not the case for the parameters of all the remaining eight phases which are assessed with new values in the present work as shown in Table 2. Fig. 1 shows the calculated Mg–Pr phase diagram compared with experimental data. Table 3 compiles the calculated invariant Mg–Pr reactions compared with the experimental data. The original heating and cooling signals tabulated in [15] are given, not just the values given in the phase diagram of [15]. The data points shown in the phase diagram for the invariant reactions are cooling data only, which are prone to supercooling. For example, the tabulated heating signal of the invariant reaction Liquid + $PrMg_3$ = Pr_5Mg_{41} is 592 °C, very close to the value of 590 °C given by Ref. [18], whereas only the cooling signal temperature, 575 °C, is found as assessed value in the phase diagram [15].

Table 2
Phase names, models (sublattice formula) and parameters of the present thermodynamic description. Gibbs energy is given in *J* per mol-formula (of sublattice definition), temperature (*T*) in Kelvin.

Liquid: (Mg, Pr)₁
$L^{0,\mathrm{Liquid}}_{\mathrm{Mg,Pr}} = -34794.8 + 9.4459\cdot T$
$L^{1,\mathrm{Liquid}}_{\mathrm{Mg,Pr}} = -18393.1$

βPr-bcc: (Mg, Pr)₁
$L^{0,\mathrm{bcc}}_{\mathrm{Mg,Pr}} = -48216.7+24.5149\cdot T$
$L^{1,\mathrm{bcc}}_{\mathrm{Mg,Pr}} = -13142.7$

αPr-dhcp: (Mg, Pr)₁
$G^{\mathrm{dhcp}}_{\mathrm{Mg}} = 5000 + G^{0,\mathrm{hcp}}_{\mathrm{Mg}}(T)$
$L^{0,\mathrm{dhcp}}_{\mathrm{Mg,Pr}} = -28466.2+16.5641\cdot T$

Pr_5Mg_{41}: $(Pr)_5$ $(Mg)_{41}$
$G^{\mathrm{Pr5Mg41}} = 5G^{0,dhcp}_{\mathrm{Pr}}(T) + 41G^{0,\mathrm{hcp}}_{\mathrm{Mg}}(T) - 371925.8 + 124.6665\cdot T$

PrMg: (Pr, Mg)₁ (Mg)₁
$G^{\mathrm{PrMg}}_{\mathrm{Mg:Mg}} = 2G^{0,\mathrm{hcp}}_{\mathrm{Mg}}(T) + 9891.5$
$G^{\mathrm{PrMg}}_{\mathrm{Pr:Mg}} = G^{0,\mathrm{dhcp}}_{\mathrm{Pr}}(T) + G^{0,\mathrm{hcp}}_{\mathrm{Mg}}(T) - 32895.1 + 8.1505\cdot T$

$PrMg_2$: $(Pr)_1$ $(Mg)_2$[a]
$G^{\mathrm{PrMg2}} = G^{0,\mathrm{dhcp}}_{\mathrm{Pr}}(T) + 2G^{0,hcp}_{\mathrm{Mg}}(T) - 32085.6 - 5.3574\cdot T$

$PrMg_3$: $(Pr)_1$ $(Mg)_3$
$G^{\mathrm{PrMg3}} = G^{0,\mathrm{dhcp}}_{\mathrm{Pr}}(T) + 3G^{0,\mathrm{hcp}}_{\mathrm{Mg}}(T) - 52672.3 + 3.9459\cdot T$

$PrMg_{12}$: $(Pr)_1(Mg)_{12}$
$G^{\mathrm{PrMg12}} = G^{0,\mathrm{dhcp}}_{\mathrm{Pr}}(T) + 12G^{0,\mathrm{hcp}}_{\mathrm{Mg}}(T) - 90417.6 + 42.6856\cdot T$

Mg-hcp: (Mg, Pr)₁
$G^{\mathrm{hcp}}_{\mathrm{Pr}} = 500 + G^{0,\mathrm{dhcp}}_{\mathrm{Pr}}(T)$
$L^{0,\mathrm{hcp}}_{\mathrm{Mg,Pr}} = -38291.9 + 31.1927\cdot T$

[a] Parameter of $PrMg_2$ taken from [4] [2008Guo].

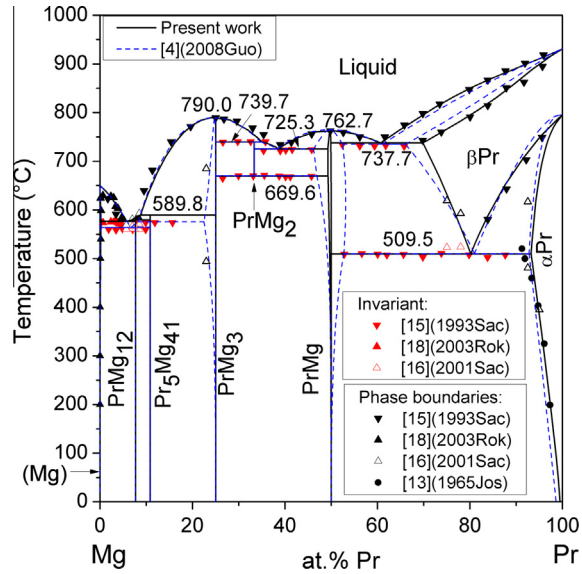

Fig. 1. Calculated Mg–Pr phase diagram compared with experimental data. Additional shorthand-notation for references, indicating year and first author, is provided in brackets.

Compositions of the invariant reactions, obtained from a graphical presentation of phase diagram data by Refs. [15,18] are not shown in Table 3 but compared with the present work by showing the individual data points in the calculated phase diagrams, Figs. 1 and 2. For example, the reported liquid composition [18] for the invariant reaction Liquid = (Mg) + $PrMg_{12}$ was extrapolated by the (Mg) liquidus and the invariant reaction temperature, 675 °C, and labeled as 4.64 at.% Pr (22 wt.% Pr) [18]. Thus, all these composition data do not provide independent validation of the calculated data.

Fig. 2 shows the calculated Mg-rich corner of the Mg–Pr phase diagram compared with experimental data. The present calculation is in good agreement with the experimental data of [18], which is considered the most accurate experimental data. It is based on the extensive work of Bochvar [14], while only few data of [15,16] are available. For the Mg-rich eutectic,

$$\mathrm{Liquid} = (\mathrm{Mg}) + \mathrm{PrMg_{12}} \tag{6}$$

the presently calculated eutectic compositions are 5.80 at.% Pr (26.31 wt.% Pr) for Liquid, 0.24 at.% Pr (1.36 wt.% Pr) for (Mg), and 7.69 at.% Pr (32.57 wt.% Pr) for stoichiometric $PrMg_{12}$ at 576.7 °C. The temperature agrees well with the accepted experimental value of 575 °C [14,18].

The most important solid solubility data of Pr in (Mg) cannot be discerned in Fig. 2, they are also experimentally determined to much lower temperature. In this dilute composition range the plot given in Fig. 3 is useful. According to Henry's law a linear relation is expected for the solubility curve in this log(at.% Pr) versus reciprocal absolute temperature plot. The linearity holds only in the composition range well below 1 at.% Pr, given in this case. Fig. 3 clearly shows that the present description agrees very well with the dedicated experimental data [18], and the improvement compared to the calculated results of [4] is recognized. It is typical that the experimental value at the lowest temperature, 200 °C, appears at composition above the assessed line, due to the experimental technique as discussed in the work on Mg–Gd [11]. For practical use in wt.%-scale the following fitting function of the solvus line in Fig. 2 is suggested:

Table 3
Invariant reactions in the Mg–Pr system.

Reaction	Temperature (°C)	Reference
Liquid = (Mg) + $PrMg_{12}$	565[a]–560[b]	[15] (1993Sac)
	575[c]	[18] (2003Rok)
	563.9	[4] (2008Guo)
	574	[5] (2012Kang)
	576.7	This work
Liquid + Pr_5Mg_{41} = $PrMg_{12}$	570[a]–565[b]	[15] (1993Sac)
	580[c]	[18] (2003Rok)
	564.3	[4] (2008Guo)
	579.6	This work
Liquid + $PrMg_3$ = Pr_5Mg_{41}	592[a]–575[b]	[15] (1993Sac)
	590[c]	[18] (2003Rok)
	576.8	[4] (2008Guo)
	593	[5] (2012Kang)
	589.8	This work
Liquid = $PrMg_3$	820[a]–790[b]	[15] (1993Sac)
	790[c]	[18] (2003Rok)
	789.6	[4] (2008Guo)
	790.0	This work
Liquid + $PrMg_3$ = $PrMg_2$	750[a]–740[b]	[15] (1993Sac)
	740[c]	[18] (2003Rok)
	739.4	[4] (2008Guo)
	738	[5] (2012Kang)
	739.7	This work
Liquid = PrMg + $PrMg_2$	730[a]–725[b]	[15] (1993Sac)
	725[c]	[18] (2003Rok)
	725.1	[4] (2008Guo)
	725.3	This work
Mg_2Pr = PrMg + $PrMg_3$	680[a]–670[b]	[15] (1993Sac)
	670	[18] (2003Rok)
	670.0	[4] (2008Guo)
	666	[5] (2012Kang)
	669.6	This work
Liquid = PrMg	770[a]–765[b]	[15] (1993Sac)
	765[c]	[18] (2003Rok)
	764.9	[4] (2008Guo)
	762.7	This work
Liquid = PrMg + βPr	745[a]–735[b]	[15] (1993Sac)
	735[c]	[18] (2003Rok)
	734.7	[4] (2008Guo)
	733	[5] (2012Kang)
	737.7	This work
βPr = PrMg + αPr	500[a]–510[b]	[15] (1993Sac)
	510[c]	[18] (2003Rok)
	510.0	[4] (2008Guo)
	529	[5] (2012Kang)
	509.5	This work

[a] Heating signals observed by Ref. [15].
[b] Cooling signals observed by Ref. [15].
[c] Evaluated experimental data from [18].

$$\log_{10}(\text{wt.\% Pr}) = 3.368 - \frac{27,501}{(T/K)} \quad (T < 849.8\ \text{K}) \tag{7}$$

For the PrMg phase a possible solid solution range of up to ±3 at.% around the ideal stoichiometry was mentioned, based on a small variation of lattice parameters [15]. However, that might be inconsistent with the thermal signals observed for these samples at the eutectics on either side in the same study [15], which should vanish for an equilibrated alloy at the edge of the single-phase region. After all, the phase boundaries of the PrMg solid solution were plotted carefully as dashed lines in [15]. The crystal structure of PrMg is undoubtedly of the cP2-CsCl ordered type. However, there is no experimental evidence at all that any order-disorder transition occurs. Disordering is not even observed in a higher order system, such as Mg–Pr–Y, where a continuous solid solution between the isostructural phases PrMg and YMg exists [4]. Therefore, we do not follow the very complicated model used for PrMg in [3,4] where an ordering amendment function is added

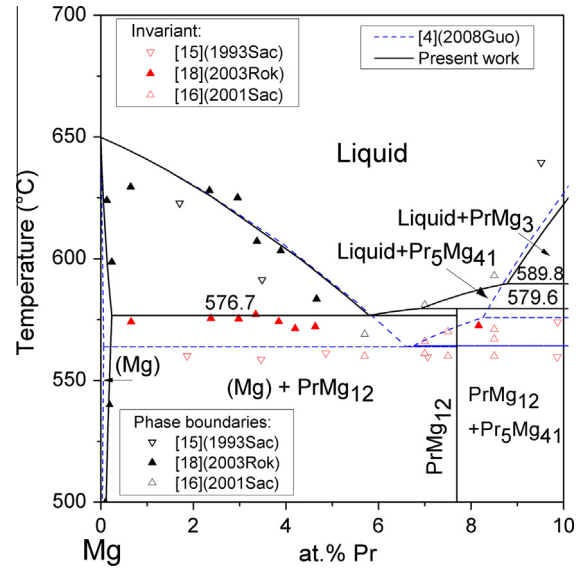

Fig. 2. Mg-rich corner of the Mg–Pr phase diagram in Fig. 1.

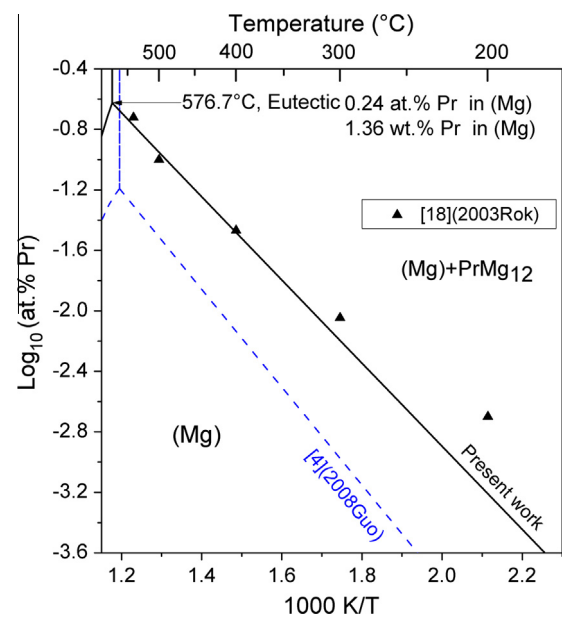

Fig. 3. Solid solubility of Pr in (Mg), comparison of calculated with experimental data.

to the description of the disordered Bcc phase, stable only as βPr in the Mg–Pr system. Since PrMg always exists as a phase separate from any Bcc phase, such as βPr, it is described here as a separate phase. A small solubility range, extending to the Mg-rich side, is proven for the YMg phase [21]. In order to maintain consistency in the multicomponent description a similar, very simple, two-sublattice model (Pr, Mg)(Mg) with no interaction parameters, given in Eq. (5), is used in the present work for the PrMg phase. The resulting very small solution range is shown in Fig. 1. It is noted that the very wide solution range obtained from the complex model in [3,4], shown as dashed line, apparently connects to the final data points of the four involved invariant reactions, even the ones at 669.6 and 509.5 °C, not indicated in the original experimental work of [15]. However, as discussed above, these data points are thermal signals which should vanish for an equilibrated alloy at the edge of the single-phase region.

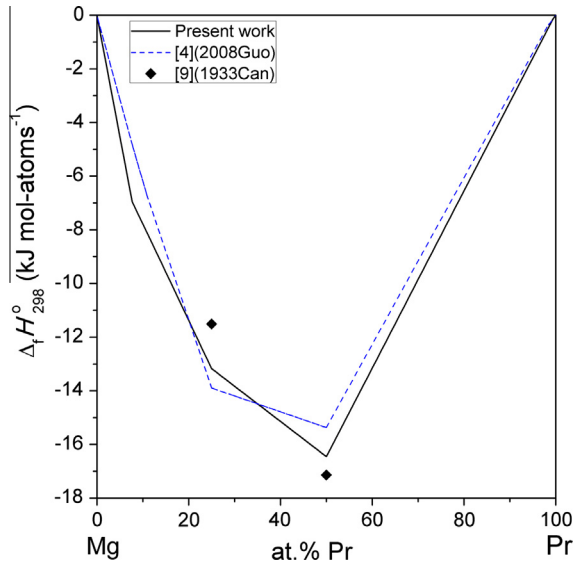

Fig. 4. Calculated standard enthalpies of formation at 298.15 K of the Mg–Pr compounds compared with scarce experimental data.

For the cF16–BiF_3 type phase $PrMg_3$ a solid solution range of up to 6 at.%, extending toward compositions richer in magnesium, has been estimated based on some lattice parameter data and metallographic appearance [15]. Subsequently a sample at 23 at.% Pr is reported to show two supporting thermal signals, however, the signal traces for that sample are not reported so it is hard to assess how significant these data are considering the very steep phase boundaries in the graphically assessed phase diagram [16]. In the assessment of [3] these two data points were exactly fitted, using a two-sublattice model (Pr, Mg)$(Mg)_3$ with two interaction parameters. However, the extended solubility at the peritectic, Liquid + $PrMg_3$ = Pr_5Mg_{41}, appears artificially large. More importantly, the peritectic temperature [3] disagrees with the accepted experimental data [18], see Table 3. In the assessment of [5] the same model (Pr, Mg)$(Mg)_3$ but, more realistically, with no interaction parameters was used. In that work only the solidus but not the solvus data point of [16] could be fitted, casting an additional doubt on that data point. The calculated peritectic temperature, 593 °C [5], is also reasonably close to the accepted 590 °C [18], see Table 3. In the present work it was decided to use the simplest model for the $PrMg_3$ phase, simple stoichiometric see Eq. (4).

The calculated Mg–Pr phase diagram in the entire composition range is shown in Fig. 1, reproducing the entity of experimental data reasonably. It is noted that in the Pr-rich region the present description is superior to both of the most recent works [4,5]. That is especially the case in comparison to the work of [5] for the solidus of βPr and for the temperature of the eutectoid βPr = PrMg + αPr, see Table 3. That has been achieved by even using less parameters in the present study, for example αPr is described in the present work as simple regular solution, compared to the subregular one in [4].

The calculated standard enthalpies of formation at 298.15 K of the Mg–Pr solid compounds are shown in Fig. 4. The experimental data shown for comparison are not well established because of the inherent problems of the acid solution calorimetry and also because the Mg starting material was of low purity [9]. The data from first-principles calculations, and even more so the values obtained from Miedema's model, shown in [5], are generally significantly less exothermic and could hardly serve as a guide line. The assessed data in [5] are even against that trend – and the experimental data – with the enthalpy of formation of $PrMg_3$ selected more exothermic than the one for PrMg. Experimentalists are encouraged to perform independent calorimetric studies on these two pure phases to obtain decisive enthalpy of formation data.

5. Conclusion

The re-assessment of the Mg–Pr system demonstrates that a simple subregular solution model for the liquid phase, available in all software packages, is sufficient for a quantitative thermodynamic description. Short-range order need not to be assumed in the Mg–Pr melt. The present work also provides a quantitative description of the phase equilibria in the important Mg-rich side. Some aspects on the Pr-rich side are also improved in the present description, compared to previous work, even though simpler models have been used. Further experiments are recommended to measure the enthalpy of formation for PrMg and/or $PrMg_3$.

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